



EcoTrace

Carbon Footprint Awareness Platform

Product Requirements Document | Challenge 3

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Date	June 15, 2026
Team	Chicha Core
Challenge	Challenge 3 — Carbon Footprint Awareness Platform
Tech Stack	Flutter · Firebase · Gemini AI · Vertex AI · Google Maps · BigQuery · Cloud Run
Target Platforms	Android · iOS · Web (Progressive Web App)
Primary Market	India (Tier 1–3 cities), expandable to SEA

1. Executive Summary

EcoTrace is an AI-powered carbon footprint awareness platform that helps individuals understand, track, and actively reduce their environmental impact. Built on Google's modern technology stack — Flutter, Firebase, Gemini AI, Vertex AI, Google Maps Platform, and BigQuery — EcoTrace delivers personalized insights, actionable nudges, and community challenges that make sustainable living simple, measurable, and motivating.

India is the world's third-largest CO₂ emitter. Transport alone accounts for over 368 million metric tons of CO₂ annually, with roadways contributing 88% of that figure. Yet individual awareness and behavior-change tools remain sparse, fragmented, and rarely localized. EcoTrace closes this gap.

Problem Statement

Most individuals have no real-time visibility into their personal carbon footprint. Existing tools are Western-centric, require manual logging, lack AI personalization, and fail to translate data into concrete daily actions. EcoTrace changes this by combining automatic activity detection, multimodal Gemini AI coaching, and gamified social challenges — all within a single cross-platform Flutter app backed by a secure Firebase + Google Cloud infrastructure.

2. Product Vision & Goals

2.1 Vision

"Make the invisible visible — turn personal carbon data into daily action and lasting habit change."

2.2 Mission

To empower every individual with clear, contextual, and culturally relevant carbon footprint data, and provide AI-driven guidance that makes sustainable choices the default — not the exception.

2.3 Key Product Goals

Goal	Success Metric	Timeline
Active user adoption	50,000 MAU within 6 months of launch	6 months
Footprint measurement accuracy	>90% agreement with ground-truth emission factor databases	At launch
Behavior change efficacy	≥15% reduction in user footprint vs. baseline after 90 days	90 days post-launch
AI coaching engagement	>60% weekly active usage of Gemini coach feature	3 months
Accessibility compliance	WCAG 2.1 AA across all screens	At launch
App performance	Cold start < 2s; API response p95 < 500ms	At launch

3. Target Users & Personas

3.1 Primary Persona — The Climate-Curious Urban Commuter

Attribute	Detail
Age	22–35
Location	Tier 1–2 Indian city (Mumbai, Pune, Bengaluru, Hyderabad)
Occupation	Working professional or student
Devices	Android-first, occasional web
Pain Point	Wants to act sustainably but doesn't know where to start or how to measure impact
Motivation	Social proof, gamification, peer comparison, cost savings tied to eco-choices

3.2 Secondary Persona — The Eco-Advocate

Attribute	Detail
Age	28–45
Location	Any metro or semi-urban area
Occupation	NGO worker, sustainability professional, educator
Pain Point	Needs data-backed insights to advocate and influence community behavior
Motivation	Community impact scores, shareable dashboards, group challenges

3.3 Tertiary Persona — The ESG-Aware Corporate Employee

Employees at companies with sustainability mandates who want to contribute personal data toward organizational ESG goals. Access to team leaderboards and office commute tracking makes EcoTrace a natural fit for B2B2C expansion.

4. Feature Specification

4.1 Carbon Footprint Dashboard (HIGH IMPACT)

Core Value Proposition
The dashboard is the heartbeat of EcoTrace — a real-time, personalized view of a user's carbon footprint across all tracked categories.

4.1.1 Emission Categories Tracked

Category	Subcategories	Data Source
Transport	Personal vehicle (fuel type, distance), public transit, auto-rickshaw, flights, electric vehicles	Google Maps Platform + manual + Google Fit
Food	Meat consumption, dairy, packaged food, food waste	Manual log + Gemini Vision (receipt/photo scan)
Energy	Home electricity (kWh), cooking fuel (LPG, PNG, induction), appliances	Manual + utility bill scan (Gemini OCR)
Shopping	Apparel, electronics, consumer goods	Manual log + bank statement parsing (optional)
Waste	Landfill waste, recycling habits, composting	Manual log

4.1.2 Dashboard UI Specifications

- Daily / Weekly / Monthly / Yearly views with animated Carbon Ring (kg CO₂ e)
- Category breakdown donut chart (Recharts-equivalent via fl_chart in Flutter)
- Trend sparkline — 30-day rolling average with color-coded threshold zones (green / amber / red)
- India national average benchmark comparison (sourced from IEA Emission Factors 2025 + Climatiq India dataset)
- Carbon Budget Meter — daily budget set by user's stated reduction goal
- Streak counter — consecutive days under target

4.2 AI-Powered Gemini Coach (HIGH IMPACT)

Google Tech Used
Gemini 1.5 Pro via Firebase AI Logic (Vertex AI in Firebase) for multimodal chat, receipt scanning, and personalized nudges. Firebase AI Logic client SDK (Dart) for Flutter integration.

4.2.1 Coach Capabilities

- Natural language Q&A: "How much CO₂ did my Mumbai–Delhi flight emit?"
- Multimodal receipt scanning: user photographs a fuel receipt or grocery bill → Gemini Vision extracts items → auto-logs emissions
- Personalized weekly insight: "Your transport emissions are 40% above your personal goal. Here are 3 swaps for next week."

- Conversational goal setting: coach negotiates realistic reduction targets based on lifestyle inputs
- Voice input via Gemini Live API for hands-free logging while commuting

4.2.2 Gemini Integration Architecture

Firebase AI Logic (Dart SDK) is used to call Gemini 1.5 Pro directly from the Flutter client with App Check security. Prompts are server-templated via Cloud Functions (Node.js) to prevent prompt injection. Structured outputs (JSON mode) ensure consistent emission log extraction from multimodal inputs.

Gemini Feature	Model	Trigger
Receipt / bill scanning	Gemini 1.5 Flash (cost-optimized)	User taps camera icon in log screen
Coach chat	Gemini 1.5 Pro	User opens Coach tab
Weekly insight generation	Gemini 1.5 Pro (batch)	Cloud Scheduler — every Sunday 7AM IST
Reduction tip personalization	Gemini 1.5 Flash	After each log entry
Voice logging	Gemini Live API	Hold mic button in log screen

4.3 Smart Activity Auto-Detection (HIGH IMPACT)

4.3.1 Google Maps Platform Integration

- Background location tracking (with explicit user consent) via Google Maps SDK for Flutter
- Activity Recognition API detects: walking, cycling, car travel, public transit
- Route carbon calculator: maps detected route to emission factors (car: ~0.192 kgCO₂ /km petrol, Metro: ~0.031 kgCO₂ /km, bus: ~0.089 kgCO₂ /km for India grid)
- Smart commute suggestion: "Your usual office commute by car emits 2.3 kg CO₂ . Taking Metro saves 1.8 kg — that's ₹45 saved too!"

4.3.2 Google Fit / Health Connect Integration

- Auto-sync walking and cycling data to offset transport emissions
- Physical activity credited against user's weekly carbon budget as a positive offset signal

4.4 Gamification & Social Layer (MEDIUM IMPACT)

4.4.1 EcoPoints System

Action	EcoPoints Awarded
Log a meal	+5 points
Choose public transit over car	+20 points
Complete a weekly challenge	+100 points
Achieve 7-day streak under target	+150 points
Refer a friend who activates account	+200 points

Plant a tree via integrated NGO partner	+500 points
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4.4.2 Challenges & Leaderboards

- Weekly City Challenges: "Pune Goes Green — Keep city transport emissions under 3 kg CO₂ for 5 days"
- Friend Duels: 2-week head-to-head emission reduction competitions
- Team Challenges for corporate cohorts
- Leaderboards segmented by city, age group, and lifestyle category

4.4.3 Achievement Badges

- "Train Tamer" — took public transit 20+ times this month
- "Plant-Based Pioneer" — zero meat week
- "Solar Saver" — first month of net-zero energy at home

4.5 Personalized Action Feed (MEDIUM IMPACT)

An algorithmic feed (powered by Vertex AI Recommendations API) surfaces the top 3 daily actions most likely to reduce the user's footprint based on their personal emission profile, local context (weather, city), and behavior history.

- "Switch your commute today — AQI in Pune is 187. Metro is healthier AND saves 1.4 kg CO₂ ."
- "You had meat 5 days this week. One plant-based day saves ~3.3 kg CO₂ ."
- "Your electricity spike on Thursday — was your AC on all day? Check our energy-saving tips."

4.6 Carbon Offset Marketplace (LOW IMPACT — Phase 2)

- Curated partner NGOs for verified tree-planting in India (linked to EcoPoints redemption)
- Renewable energy certificate (REC) purchases via Google Pay integration
- Transparency dashboard: user sees exactly where their offset investment went

4.7 Accessibility Features (HIGH IMPACT)

- WCAG 2.1 AA compliance across all screens
- Dynamic text scaling support (Flutter TextScaleFactor)
- Screen reader support (TalkBack / VoiceOver) — all interactive elements semantically labeled
- High contrast mode toggle
- Color-blind safe palette (avoids red/green as the only differentiator; uses icons + labels)
- Multilingual: English, Hindi, Marathi at launch (Google Cloud Translation API for future languages)
- Offline mode: last 7 days of data cached via Hive local storage; sync on reconnect

5. Google Technology Stack

Design Principle

Every layer of EcoTrace's stack uses Google-first technologies — ensuring deep integration, reduced operational overhead, and eligibility for Google for Startups / GCP credits.

Layer	Google Technology	Purpose	Justification
Frontend	Flutter 3.x (Dart)	Cross-platform iOS/Android/Web from single codebase	Google I/O 2025 showcased Flutter + Firebase + Gemini as the gold standard for multiplatform AI apps
Auth	Firebase Authentication	Email/Google SSO/Phone OTP	Built-in security, Flutter SDK, free tier generous
Database	Cloud Firestore	Real-time emission logs, user profiles, social features	Real-time sync, offline persistence, security rules
AI / LLM	Gemini 1.5 Pro & Flash via Firebase AI Logic	Coach, receipt scan, weekly insights, voice logging	Direct Dart SDK, App Check security, multimodal support
AI Recommendations	Vertex AI Recommendations API	Personalized action feed	Collaborative filtering on emission profiles
Data Warehouse	BigQuery	Aggregated analytics, city-level benchmarks, admin dashboards	Petabyte-scale, SQL-familiar, BigQuery ML for trend models
Backend Logic	Cloud Functions (Node.js)	Prompt templating, scheduled batch jobs, webhook handlers	Serverless, auto-scaling, Firebase-native
Scheduling	Cloud Scheduler	Weekly insight generation, challenge refresh, leaderboard resets	Cron-based, reliable, GCP-native
Maps & Location	Google Maps Platform (Maps SDK, Routes API, Places API)	Activity detection, commute carbon calc, location context	Best-in-class India map data; routes API for emission calculation
Health Data	Health Connect (Android) / Google Fit REST API	Auto-sync walking and cycling distance	Native Android integration, user consent-gated
Notifications	Firebase Cloud Messaging (FCM)	Daily nudges, streak alerts, challenge updates	Free, reliable, cross-platform push
Crash Reporting	Firebase Crashlytics	Real-time crash detection and stack traces	Zero config with Flutter Firebase
Analytics	Firebase Analytics + Google Analytics 4	User funnel, feature adoption, A/B test tracking	Free, GA4 integration for unified reporting

A/B Testing	Firebase Remote Config + A/B Testing	Test nudge message variants, UI flows, coach prompts	Server-driven config, no app update needed
App Security	Firebase App Check (Play Integrity / DeviceCheck)	Prevent API abuse, protect Gemini API quota	Required for Firebase AI Logic production use
Storage	Firebase Storage	Receipt images, profile photos, exported reports	CDN-backed, secure rules, Flutter SDK
Hosting / Web	Firebase App Hosting	PWA hosting with CDN	GA in 2025, native Flutter Web support
CI/CD	Cloud Build + Artifact Registry	Automated build, test, deploy pipeline	GCP-native, integrates with GitHub
Monitoring	Cloud Monitoring + Cloud Logging	API latency, error rate SLOs, quota alerts	Full-stack observability
Translation	Cloud Translation API	On-demand multilingual content localization	Neural MT, 100+ languages, pay-per-use

6. System Architecture

6.1 High-Level Architecture

EcoTrace follows a serverless, event-driven architecture with a Flutter frontend communicating directly with Firebase services for real-time data and Cloud Functions for business logic. Gemini AI calls are made securely via Firebase AI Logic with App Check enforcement.

Architecture Layers

CLIENT LAYER: Flutter app (iOS/Android/Web) with local Hive cache FIREBASE LAYER: Auth · Firestore · Storage · FCM · App Check · Remote Config · Crashlytics AI LAYER: Gemini 1.5 Pro/Flash via Firebase AI Logic · Vertex AI Recommendations BACKEND LAYER: Cloud Functions (Node.js) · Cloud Scheduler · Cloud Run (for heavy batch) DATA LAYER: BigQuery (analytics warehouse) · Firestore (operational store) INFRA LAYER: Cloud Build · Artifact Registry · Cloud Monitoring · Cloud Logging

6.2 Data Flow — Emission Log Entry

1. User logs activity (transport, food, energy) via Flutter UI or Gemini voice/photo
2. Client validates input locally, writes to Firestore `/users/{uid}/emissions/{date}/{entryId}`
3. Firestore trigger fires Cloud Function → calculates CO₂ e using emission factor lookup (Firestore `/emissionFactors` collection, sourced from IEA 2025 + Climatiq India)
4. Calculated emission written back to Firestore; client listens via StreamBuilder → UI updates in real-time
5. Batch export to BigQuery via Cloud Function on daily schedule for analytics

6.3 Data Flow — Gemini Coach

6. User sends message or photo in Coach tab
7. Flutter calls Firebase AI Logic Dart SDK with user context (last 7-day emission summary injected as system context)
8. App Check token verified by Firebase before forwarding to Gemini API
9. Gemini response streamed back to client (streaming text support in Firebase AI Logic SDK)
10. Extracted structured data (e.g., emission values from scanned receipt) triggers auto-log flow

6.4 Security Architecture

- All Firestore collections protected by fine-grained Security Rules: users can only read/write their own data
- Firebase App Check (Google Play Integrity on Android, App Attest on iOS) required for all Firebase service calls
- No raw API keys in Flutter source — all secrets in Cloud Secret Manager, accessed via Cloud Functions only
- PII minimization: location data hashed to 5km grid squares for analytics; raw GPS never stored
- GDPR / DPDP Act 2023 (India) compliant: explicit consent flows, data export, account deletion
- Cloud Armor WAF on Cloud Run endpoints for DDoS protection

7. Emission Factor Database

7.1 Data Sources

Accurate CO₂ e calculation is EcoTrace's scientific foundation. We use a curated, version-controlled emission factor database stored in Firestore, sourced from:

Source	Coverage	Update Frequency
IEA Emission Factors 2025	Electricity grid intensity by country/region (India: 0.82 kgCO ₂ /kWh)	Annual
Government of India — Central Electricity Authority	State-level grid emission factors	Annual
Climatiq India Dataset	Transport, food, goods emission factors for India	Quarterly
IPCC AR6 (2022)	Global warming potentials (GWP100) for CH ₄ , N ₂ O	Per IPCC cycle
AI4Bharat / ResearchGate — Indian food carbon data	Food category emission factors (kgCO ₂ e/kg) for Indian diet	Annual review

7.2 Key Emission Factors (India, 2025)

Activity	Emission Factor	Unit
Electricity consumption (India grid avg)	0.82	kgCO ₂ e/kWh
Petrol car (personal)	0.192	kgCO ₂ e/km
Diesel car (personal)	0.171	kgCO ₂ e/km
CNG auto-rickshaw	0.089	kgCO ₂ e/km
Urban metro / electric rail	0.031	kgCO ₂ e/km
Diesel bus	0.089	kgCO ₂ e/km (per passenger)
Domestic flight (economy)	0.255	kgCO ₂ e/km (per passenger)
Beef (1 kg)	27.0	kgCO ₂ e/kg
Chicken (1 kg)	6.9	kgCO ₂ e/kg
Rice (1 kg)	2.7	kgCO ₂ e/kg
LPG cooking (1 cylinder 14.2kg)	38.6	kgCO ₂ e/cylinder

8. User Journey & Screens

8.1 Onboarding Flow

- 11. Splash → App intro (3 swipeable value prop cards)
- 12. Sign up via Google SSO or phone OTP (Firebase Auth)
- 13. Lifestyle Quiz (5 questions): primary transport mode, diet type, home energy source, city, household size → generates baseline footprint estimate instantly
- 14. Goal Setting: Gemini coach introduces personalized 3-month reduction goal
- 15. Notification permission request with clear value framing
- 16. Home Dashboard — fully populated with baseline data

8.2 Core Screen Inventory

Screen	Key Components	Google Tech
Home Dashboard	Carbon Ring, category bars, streak, today's top action	Firestore real-time listener
Log Activity	Quick-log buttons, camera (Gemini scan), voice input	Firebase AI Logic, Gemini Live API
Gemini Coach	Chat interface, message streaming, attachment support	Firebase AI Logic Dart SDK
Trends & Insights	Weekly/monthly charts, peer comparison, AI-written insight card	BigQuery data, Gemini batch insights
Challenges & Social	Active challenges, friend leaderboard, city ranking	Firestore, FCM
Action Feed	Personalized 3-card daily feed, impact preview	Vertex AI Recommendations
Profile & Settings	Goals, notification prefs, data export, account deletion	Firebase Auth, Firestore
Offset Marketplace	NGO partner cards, EcoPoints redemption, Google Pay	Cloud Functions, Google Pay API

9. Technical Requirements & Code Quality

9.1 Flutter Architecture

EcoTrace uses a clean BLoC (Business Logic Component) architecture for state management, ensuring separation of concerns and testability.

- Presentation Layer: Flutter Widgets (stateless preferred) + BLoC state
- Domain Layer: Pure Dart use cases, entity models, repository interfaces
- Data Layer: Firebase Firestore repositories, Gemini AI Logic service, Google Maps service
- DI: get_it + injectable for dependency injection
- Routing: GoRouter for declarative navigation

9.2 Code Quality Standards

Standard	Requirement
Linting	flutter_lints + custom analysis_options.yaml with strict null safety
Code coverage	≥ 80% unit test coverage on domain layer use cases
Integration tests	Flutter integration_test package covering all critical user flows
Widget tests	All shared UI components have widget test snapshots
CI enforcement	Cloud Build fails PR merge on lint errors, failing tests, or coverage drop below threshold
Documentation	All public APIs documented with dartdoc; Architecture Decision Records (ADRs) in /docs/adr/
Code review	Minimum 1 peer review + automated Gemini Code Assist review on all PRs

9.3 Performance Requirements

Metric	Target	Measurement Method
Cold start time (Android)	< 2 seconds	Firebase Performance Monitoring
Dashboard render time	< 500ms on mid-range device	Flutter DevTools frame rendering
Firestore query p95	< 300ms	Cloud Monitoring custom metric
Gemini coach response (first token)	< 1.5s	Client-side timer, Firebase Performance
App size (Android APK)	< 25MB	CI build artifact size check
Offline capability	Full read access for last 7 days without internet	Integration test with airplane mode
Battery impact	< 2% per hour with background location on	Android battery historian

9.4 Security Implementation

- Firebase Security Rules: unit-tested with @firebase/rules-unit-testing in CI pipeline
- App Check: enforced in production, bypassed only in emulator-flagged CI environments
- Input sanitization: all user text inputs sanitized before Gemini prompt injection
- Rate limiting: Cloud Functions enforce per-user API call limits via Firestore counters
- Data encryption: Firestore at-rest encryption (Google-managed keys); option for CMEK for enterprise tier
- Dependency auditing: npm audit + dart pub outdated run in CI on every merge

10. Testing Strategy

10.1 Testing Pyramid

Test Type	Scope	Tool	Coverage Target
Unit Tests	Domain use cases, emission calculators, data parsers	flutter_test	≥ 80%
Widget Tests	All shared UI components, form validation, state rendering	flutter_test + golden_toolkit	≥ 70% of widgets
Integration Tests	End-to-end user flows (onboarding, log activity, view dashboard)	Flutter integration_test + Firebase Emulator Suite	10 critical flows
Security Rule Tests	Firestore security rules for all collections	@firebase/rules-unit-testing (Node.js)	100% rule coverage
API Contract Tests	Cloud Functions HTTP endpoints	Jest + supertest	All endpoints
Performance Tests	Render time, Firestore query latency, Gemini response time	Firebase Performance Monitoring + Dart benchmarks	SLOs per §9.3
Accessibility Tests	Screen reader, contrast ratios, touch target sizes	Flutter semantics tests + Lighthouse (web)	WCAG 2.1 AA
Load Tests	Cloud Functions + Firestore under concurrent user load	k6 + Google Cloud Load Testing	1,000 concurrent users

10.2 Firebase Emulator Suite

All local development and CI test runs use the Firebase Local Emulator Suite (Auth, Firestore, Functions, Storage, FCM) to eliminate flaky tests caused by network dependency and avoid production data pollution.

10.3 Gemini AI Testing

- Prompt unit tests: deterministic assertion tests on JSON-mode structured outputs
- Adversarial testing: prompt injection attempts, malformed receipt images, edge-case dietary inputs
- Golden response tests: weekly insight outputs evaluated against quality rubric (accuracy, tone, actionability) by human raters monthly

11. Accessibility & Inclusive Design

11.1 WCAG 2.1 AA Compliance Matrix

WCAG Criterion	Implementation in EcoTrace
1.1.1 Non-text Content	All icons have Semantics labels; chart segments have text equivalents
1.3.1 Info and Relationships	Flutter Semantics tree correctly reflects heading hierarchy
1.4.3 Contrast (Minimum)	All text meets 4.5:1 contrast ratio; verified with Colour Contrast Analyser
1.4.4 Resize Text	App respects system font size up to 200% without horizontal scroll
2.1.1 Keyboard	Web version fully keyboard navigable; logical tab order
2.4.7 Focus Visible	Custom FocusNode with visible ring on all interactive elements
3.1.1 Language of Page	lang attribute set; Semantics.textDirection for RTL-ready strings
4.1.2 Name, Role, Value	All custom widgets use ExcludeSemantics or MergeSemantics correctly

11.2 Localization

- Flutter l10n (ARB files) for English, Hindi (हिन्दी), Marathi (मराठी) at launch
- RTL layout support via Directionality widget for future Arabic/Urdu expansion
- Locale-aware number formatting (e.g., Indian numbering system: 1,00,000 not 100,000)
- Cloud Translation API for admin-added community content beyond core strings

11.3 Offline & Low-Connectivity Design

- Hive local database caches 7 days of emission logs, all emission factor tables, and last AI insight
- Firestore's built-in offline persistence handles real-time sync gracefully
- Graceful degradation: Gemini features show queued state with "Will sync when online" message
- Optimized for 3G: image assets lazy-loaded, API payloads compressed (gzip), Flutter tree-shaken

12. Data Privacy & Compliance

Requirement	Implementation
India DPDP Act 2023	Consent manager on first launch; purpose-limited data collection; Data Principal rights dashboard (view, correct, delete)
GDPR (for international users)	DPA-compliant Firebase processing agreement; data residency options via Firebase multi-region
Location Data	Raw GPS never stored; hashed to 5km grid; background location requires explicit opt-in with clear value explanation
Children's Privacy	Age gate at onboarding; users under 13 blocked from account creation
Data Retention	Active users: indefinite; deleted accounts purged within 30 days (Firestore TTL + Cloud Functions)
Third-Party Sharing	No data sold or shared with advertisers; only aggregated, anonymized city-level data shared with NGO partners
AI Training	User data not used to train Gemini models (Firebase AI Logic API terms)
Right to Portability	One-click JSON export of all personal emission data via Firebase Extensions (Export Collections to BigQuery)

13. Efficiency & Resource Optimization

13.1 Gemini Cost Optimization

- Route coach queries through Gemini Flash (cheaper) for simple factual Q&A; escalate to Pro only for complex reasoning
- Cache weekly insight results in Firestore for 7 days — avoid regenerating for same user data
- Batch weekly insight generation using Cloud Scheduler (off-peak hours, 2AM IST Sunday) to leverage lower inference latency
- Implement input token budgets: inject only last 7-day emission summary (~500 tokens) not full history

13.2 Firestore Optimization

- Composite indexes pre-built for all common query patterns (user + date range, category + date)
- Pagination with startAfter cursors on all list queries — no unbounded reads
- Emission log aggregation: daily summary documents pre-computed by Cloud Function trigger → dashboard reads one document, not 50 log entries
- Firestore TTL policy on raw GPS location documents: 48-hour auto-deletion

13.3 Flutter App Efficiency

- Flutter tree-shaking removes unused icon glyphs and fonts (reduces APK by ~2MB)
- Lazy loading: Action Feed and Offset Marketplace loaded on demand, not at app startup
- Image optimization: Firebase Storage serves WebP via Cloud CDN; Flutter `CachedNetworkImage` with disk cache
- BLoC state management prevents unnecessary widget rebuilds via precise event/state modeling

14. Product Roadmap

Phase	Timeline	Key Deliverables
Phase 0 — MVP	Month 1–2	Core logging (transport, food, energy), dashboard, Gemini coach (text), Firebase backend, Google Maps commute tracking, onboarding
Phase 1 — Engagement	Month 3–4	Gamification (EcoPoints, streaks, badges), social challenges, FCM notifications, multimodal Gemini (receipt scan), multilingual (Hindi)
Phase 2 — Intelligence	Month 5–6	Vertex AI Recommendations (action feed), BigQuery analytics dashboard, Gemini voice logging, Marathi localization, WCAG audit
Phase 3 — Scale	Month 7–9	Corporate B2B2C module, Offset Marketplace, city-level public dashboards, Cloud Armor hardening, load testing at 50K MAU
Phase 4 — Expansion	Month 10–12	SEA market localization, Carbon API for third-party integrations, On-device Gemma 4 for offline AI coaching

15. Success Metrics & Evaluation Criteria

15.1 Alignment with Evaluation Focus Areas

Evaluation Area	Impact Level	EcoTrace Implementation	Metric
Code Quality	High	BLoC architecture, strict linting, ADRs, Gemini Code Assist PR reviews	0 lint errors in CI; architecture review score
Security	High	App Check, Security Rules tests, no raw keys, DPDP compliance, WAF	0 critical vulnerabilities in security audit; 100% rule test coverage
Efficiency	Medium	Gemini Flash/Pro routing, Firestore aggregation, Flutter tree-shaking, batch scheduling	p95 Firestore < 300ms; Gemini cost < \$0.01/user/month at 50K MAU
Testing	Medium	4-layer test pyramid, emulator suite, golden AI tests, load tests	≥80% unit coverage; 10 integration flows passing; 1000 concurrent load test pass
Accessibility	High	WCAG 2.1 AA, TalkBack/VoiceOver, multilingual, offline, color-blind safe	Lighthouse accessibility score ≥ 95; 0 critical a11y issues

15.2 North Star Metrics

- Monthly Active Users (MAU): 50,000 at 6 months
- Average CO₂ reduction per active user vs. baseline: ≥ 15% at 90 days
- Day-30 retention: ≥ 35%
- Gemini coach weekly engagement: ≥ 60% of MAU
- Net Promoter Score (NPS): ≥ 45

16. Risks & Mitigations

Risk	Severity	Mitigation
Location permission rejection by users (limits auto-detection)	High	Design for manual log as equally first-class UX; value-frame location as 'saves you typing 5x per day'
Gemini API latency spikes degrading Coach UX	Medium	Streaming responses mask latency; fallback to cached tips if API >3s
Emission factor data staleness	Medium	Versioned factor database with Cloud Scheduler-triggered update checks; user notification on major factor changes
Firebase cost overrun at scale	Medium	Firestore read quota alerts in Cloud Monitoring; BigQuery slot reservations for analytics; Gemini cost cap via Firebase budget alerts
User data accuracy (self-reported diet/shopping)	Low	Gemini multimodal scanning reduces manual entry; confidence scoring on AI-extracted values
Competitor with Google backing launches similar product	High	Speed to market + India-first localization + community depth as moat; B2B2C corporate channel for defensibility

17. Appendix

17.1 API Reference Summary

API	Endpoint / SDK	Auth Method
Firebase Auth	Firebase Flutter SDK	Firebase App Check + ID Token
Cloud Firestore	cloud_firestore (Dart)	Firebase Security Rules + App Check
Firebase AI Logic (Gemini)	firebase_ai (Dart SDK)	App Check + Firebase Auth
Google Maps SDK	google_maps_flutter	API Key (restricted by bundle ID)
Cloud Functions	HTTPS Callable Functions	Firebase Auth ID Token
BigQuery	BigQuery REST API (server-side only)	Service Account (Cloud Functions)
Vertex AI Recommendations	Vertex AI REST API (server-side)	Service Account
Cloud Translation	Translation REST API (server-side)	Service Account
Firebase Cloud Messaging	firebase_messaging (Dart)	FCM Server Key (server-side)
Health Connect	health (Flutter package)	User Consent (Android)

17.2 Emission Category Data Model (Firestore)

Collection: /users/{uid}/emissions/{YYYY-MM-DD}/entries/{entryId}

Field	Type	Description
category	String	transport food energy shopping waste
subcategory	String	e.g., 'petrol_car', 'beef', 'lpg_cooking'
value	Number	Quantity in native unit (km, kg, kWh)
unit	String	km kg kWh trips items
co2e_kg	Number	Calculated CO ₂ equivalent in kg
factor_version	String	Version ID of emission factor used (e.g., 'IEA_2025_IN')
source	String	manual gemini_scan maps_auto google_fit
confidence	Number	0.0–1.0 (AI-extracted entries include confidence score)
created_at	Timestamp	Firestore server timestamp
location_grid	String	5km grid hash (transport entries only, for city analytics)

